

APS110 – Lecture 7

See hand written notes for drawings.

FCC – Face-Centered Cubic

- Remember, this cube is the unit cell of the larger structure
- The corners of the cube provide one full sphere (remember, we're only concerned with what's inside the cube)
- The faces of the cube provide 3 full spheres
- In total (n_{FCC}) we have 4 full spheres
- Different properties in different directions
- Ex. we can calculate *theoretical density*
 - o Density (ρ) = mass per unit volume (m/V)
 - o A = molar mass (or, atomic weight)
 - Atomic weight is the more antiquated term, it's discouraged
 - o n = the number of atoms in the unit cell $\rightarrow$ for FCC, this is 4
 - o V_c = cubic volume
 - It happens that the periodic table is filled with things that have cubic symmetry in their crystal structure (...lucky us!)
 - o N_A = Avogadro's number ($6.022 \times 10^{23} \text{ mol}$)
 - o Note that g/m^3 is not a standard unit of density $\rightarrow$ we'd often need to convert to either kg/m^3 or g/cm^3
- Ex. density of aluminum
 - o R_{Al} = atomic radius of aluminum
 - Note: R will be used to denote atomic radius, r will be used to denote interatomic spacing
 - o In the question, it will always indicate the elements that have FCC crystal structure (so you will know $n=4$)
 - o With FCC, it's a cube, so there's one lattice parameter that defines the dimensions $\rightarrow a$ (length of any one side)
 - You can see from the diagram, that the hypotenuse of the face side is made up of 4 atomic radii ($4R$)
 - So for FCC, $a = 2\sqrt{2} R$

Another cubic structure $\rightarrow$ ceramic structure, common

- NaCl – “rock salt”
 - o There are a lot of structures that have the “rock salt” crystal structure
 - o With ceramic structures, they're named after the *most common* material with that structure
- With ceramic structures, there is **more than one atom type**
 - o With NaCl, there is both Na and Cl (obviously)
- NaCl structure is also **cubic**

- Usually, anions tend to be larger and so when drawing the structure, we tend to draw the anions first
- Stoichiometry → because there is one Na for every Cl, we now have to fill in with an equal number of cations (Na)
- Edges of the cube generate $\frac{1}{4}$ of a sphere
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